Transport networks传输网

- Basic concepts
- Labeling algorithm
- The Max flow Min Cut Theorem

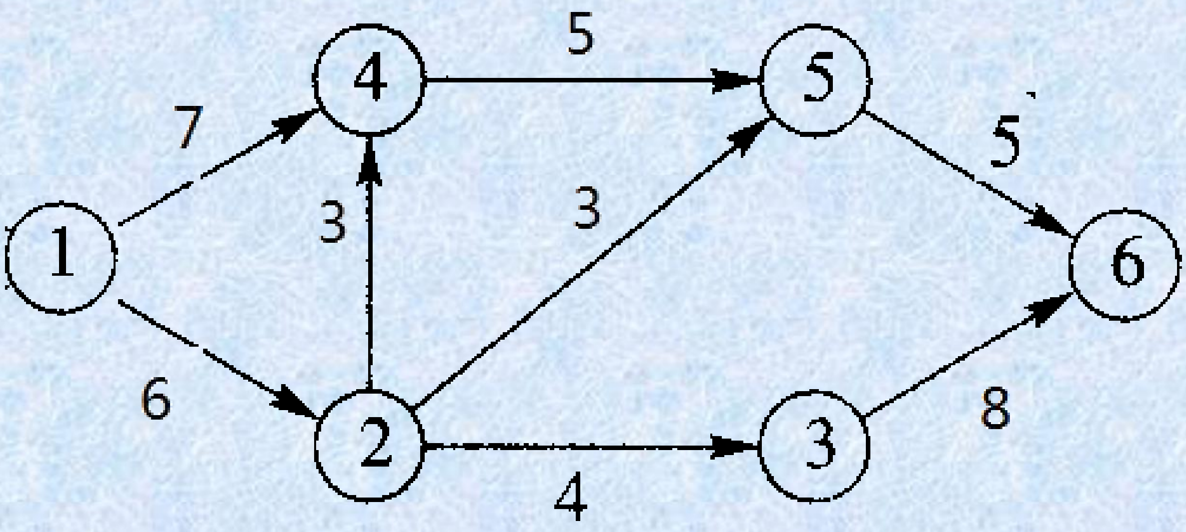
P297 8.4

Directed graph (Digraph)

- An important use of labeled digraphs is to model what are commonly called transport networks.
- The label on an edge represents the maximum flow that can be passed through that edge and is called the *capacity*（容量）of the edge. Many situations can be modeled in this way.

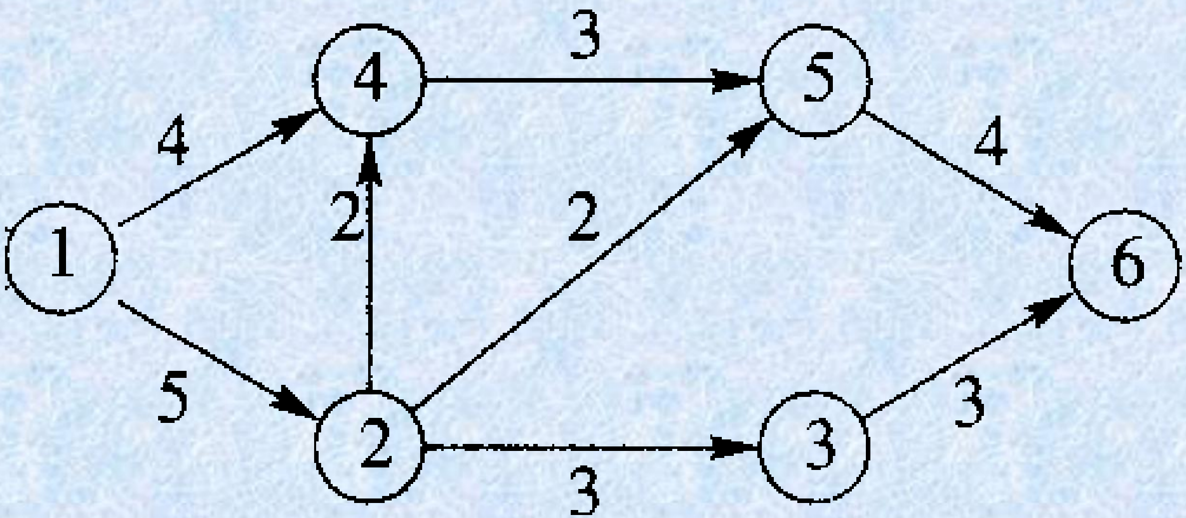
Example

- Figure below might as easily represent an oil pipeline, a highway system, a communications network, or an electric power grid.



Example

- The vertices of a network are usually called nodes and may denote pumping stations, shipping depots, relay stations, or highway interchanges.



Transport network

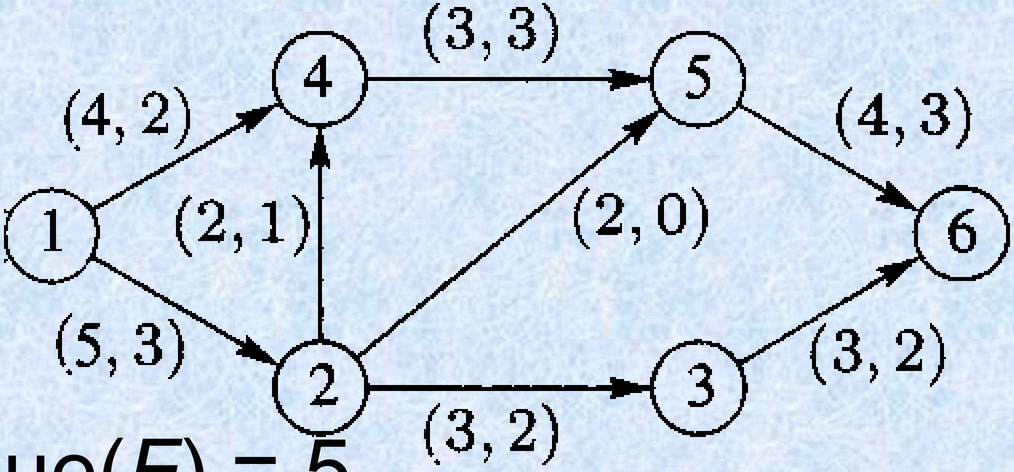
- More formally, a *transport network*, or a *network*, is a connected digraph N with the following properties:
  - There is a unique node, the *source（源端）*, that has in-degree 0. We generally label the source node I.
  - There is a unique node, the *sink（宿端）*, that has out-degree 0. If N has n nodes, we generally label the sink as node n.
  - The graph N is labeled. The label, C_{ij}, on edge (i, j) is a nonnegative number called the *capacity* of the edge.

Flows

- Mathematically, a *flow*（流量） in a network N is a function that assigns to each edge (i, j) of N a nonnegative number F_{ij} that does not exceed C_{ij}.
  - *Conservation of flow*（流量守恒）
  - *Value of the flow*

Flows

- We can represent a flow *F* by labeling each edge (*i, j*) with the pair (C_{ij}, F_{ij})



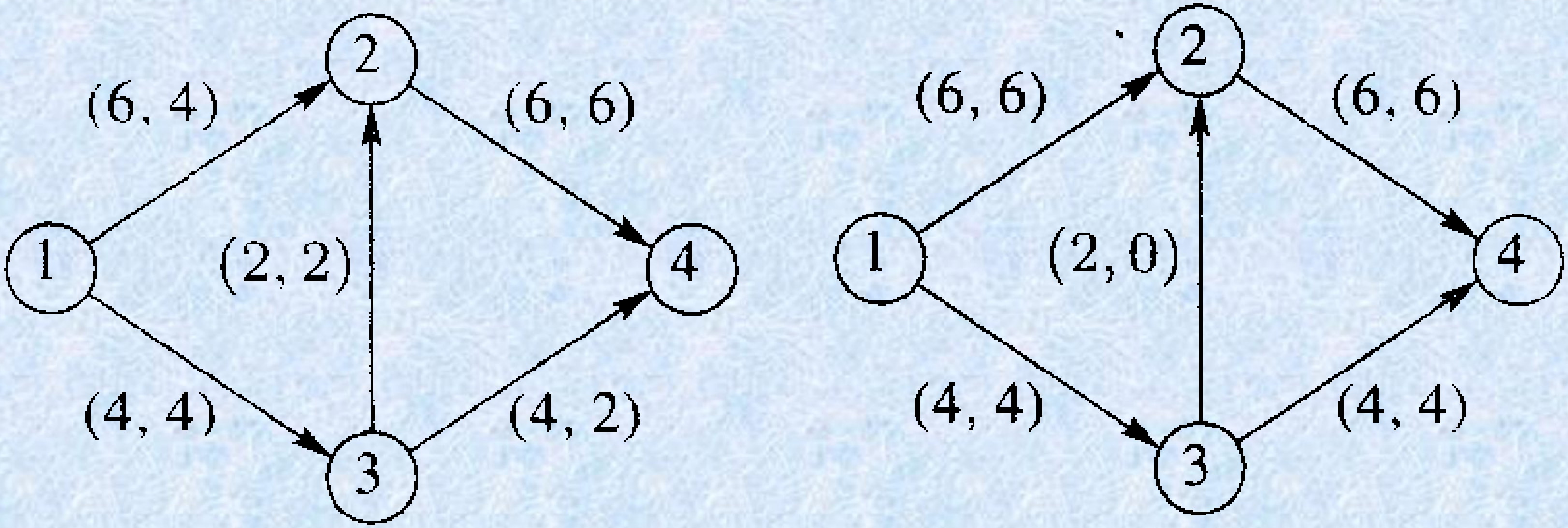
- Here value(*F*) = 5

Maximum Flows（最大流量）

- For any network an important problem is to determine the maximum value of a flow through the network and to describe a flow that has the maximum value.
- For obvious reasons this is commonly referred to the *maximum flow problem*.

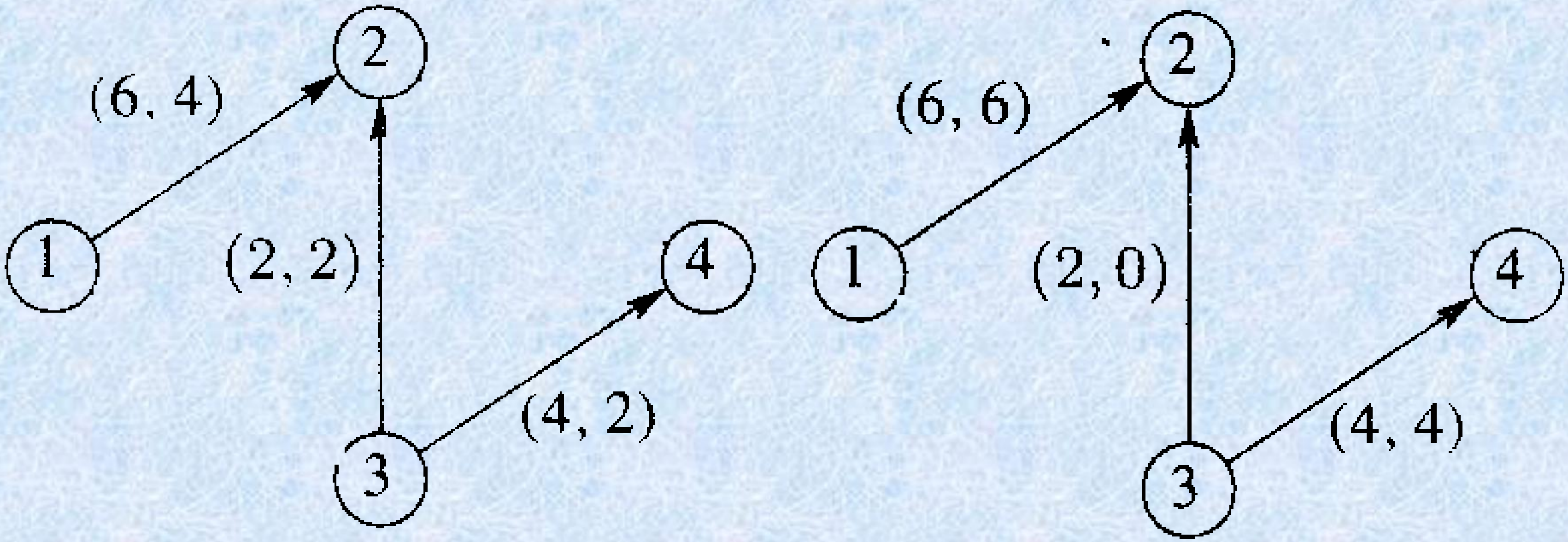
Example 2

- Even for a small network, we need a systematic procedure for solving the maximum flow problem.



(a) (b)

Virtual Flow



- Let N be a network and let G be the symmetric closure of N.
  - Choose a path in G and let an edge (i,j) in this path.
  - If $(i,j)\in N$ and $e_{ij}=C_{ij}-F_{ij}>0$,then we say this edge has positive excess capacity.
  - If (i,j) is not an edge of N then we are traveling this edge in the wrong direction. $E_{ij}=F_{ji}$ if $F_{ji}>0$. Increasing flow through edge(i,j) will have the effect of reducing F_{ji}.

LABELING ALGORITHM

- The algorithm we present is due to Ford and Fulkerson and is often called the *labeling algorithm*（*标记算法*）. The labeling referred to is an additional labeling of nodes. We have used integer capacities for simplicity, but Ford and Fulkerson show that this algorithm will stop in a finite number of steps if the capacities are rational numbers.

LABELING ALGORITHM

- Let *N* be a network with *n* nodes and *G* be the symmetric closure of *N*. All edges and paths used are in *G*.
  - Begin with all flows set to 0.
  - As we proceed, it will be convenient to track the excess capacities in the edges and how they change rather than tracking the increasing flows.
  - When the algorithm terminates, it is easy to find the maximum flow from the final excess capacities.

Step 1

- Let N_1 be the set of all nodes connected to the source by an edge with positive excess capacity.
- Label each j in N_1 with $[E_j, \mathrm{I}]$, where E_j is the excess capacity e_{Ij} of edge (I, j).
- The I in the label indicates that j is connected to the source, node I.

Step 2

- Let node j in N_1 be the node with smallest node number and let N_2(j) be the set of all unlabeled nodes, other than the source, that are joined to node j and have positive excess capacity.
- Suppose that node k is in N_2(j) and (j, k) is the edge with positive excess capacity. Label node k with [E_k, j], where E_k is the minimum of E_j and the excess capacity e_{jk} of edge (j, k).
- When all the nodes in $N_2(j)$ are labeled in this way, repeat this process for the other nodes in N_1. Let

$$N_2 = \bigcup_{j \in N_1} N_2(j)$$

Step 3

- Repeat Step 2, labeling all previously unlabeled nodes N_3 that can be reached from a node in N_2 by an edge having positive excess capacity.
- Continue this process forming sets N_4, N_5. . . . until after a finite number of steps either
  - (i) the sink has not been labeled and no other nodes can be labeled. It can happen that no nodes have been labeled; remember that the source is not labeled. or
  - (ii) the sink has been labeled.

Step 4

- In case (i), the algorithm terminates and the total flow then is a maximum flow.

Step 5

- In case (ii) the sink, node n, has been labeled with [E_n, m] where E_n is the amount of extra flow that can be made to reach the sink through a path π.

Step 5

- We examine π in reverse order.
  - If edge $(i, j) \in N$, then we increase the flow in (I, j) by E_n and decrease the excess capacity e_{ij} by the same amount. Simultaneously, we increase the excess capacity of the (virtual) edge (j, i) by E_n since there is that much more flow in (i, j) to reverse.
  - If, on the other hand, $(i, j) \notin N$, we decrease the flow in (j, i) by E_n and increase its excess capacity by E_n. We simultaneously decrease the excess capacity in (i, j) by the same amount, since there is less flow in (i, j) to reverse.
  - We now have a new flow that is E_n units greater than before and we return to Step 1.